ORFhunteR: automatic identification and annotation of open reading frames in human mRNA molecules

Vasily V. Grinev[1], Mikalai M. Yatskou[2], Victor V. Skakun[2], Maryna Chepeleva[2,3] and Petr V. Nazarov[3]

[1] Department of Genetics, Belarusian State University, Minsk, Belarus
[2] Department of Systems Analysis and Computer Modelling, Belarusian State University, Minsk, Belarus
[3] Multiomics Data Science Research Group, Quantitative Biology Unit, Luxembourg Institute of Health, Strassen, Luxembourg

bioinformatics.rfct.bio.bsu@gmail.com, maryna.chepeleva@gmail.com

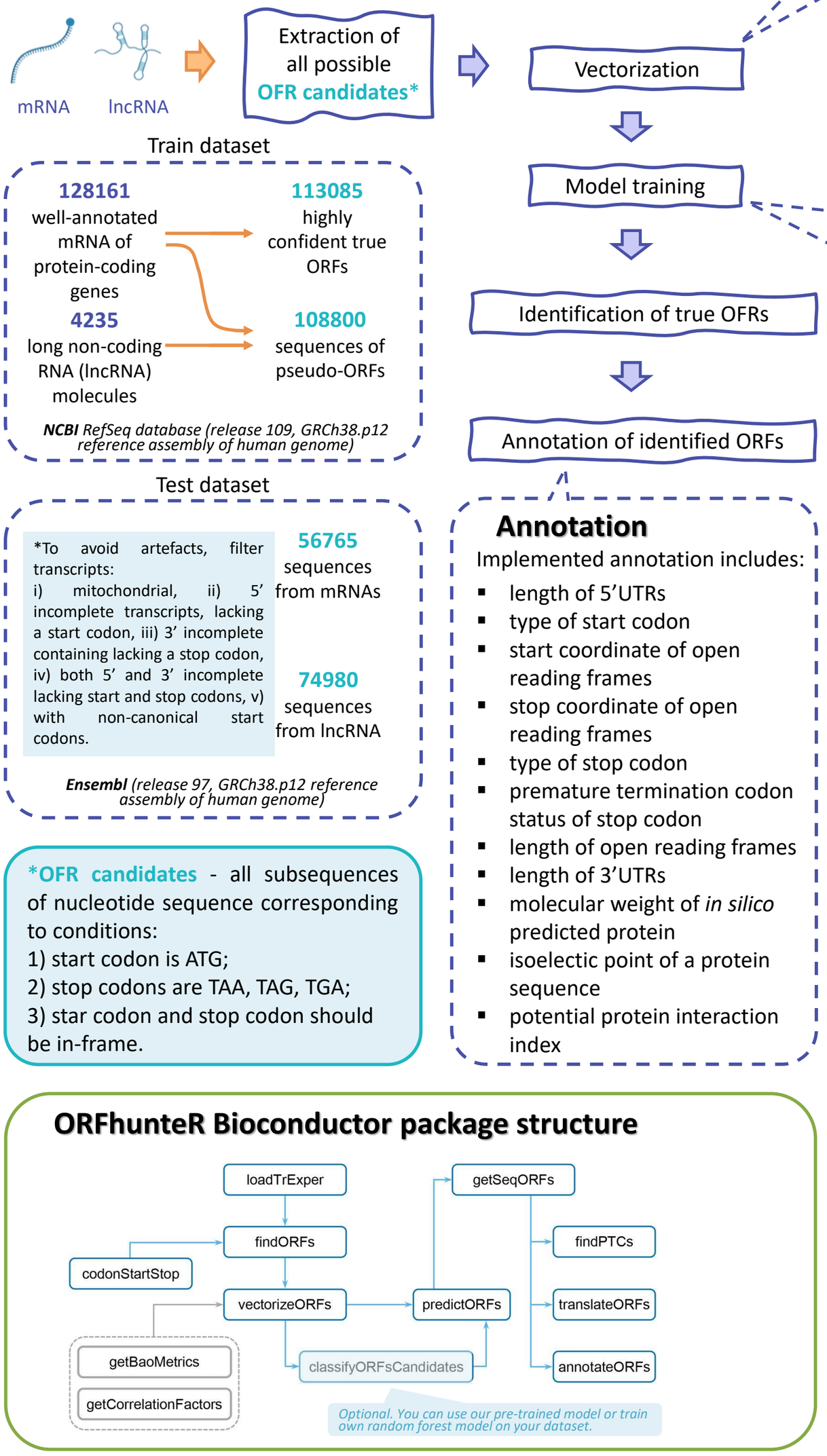
https://orfhunter.bsu.by/

Introduction

Modern methods of whole transcriptome sequencing accurately recover nucleotide sequences of RNA molecules present in cells and allow to determine their quantitative abundances [1]. The coding potential of such molecules can be estimated using algorithms that find open reading frames (ORF). However, these algorithms show limited accuracy, are intended for single-molecule analysis and do not allow selecting proper ORFs in the case of long mRNAs containing multiple ORF candidates.

Here we propose ORFhunteR – computational tool that performs an automatic identification of true ORFs among large set of human mRNA molecules based on vectorization of nucleotide sequences and random forest classifier [2].

Material and Methods



Vectorization

(transforming the sequence into the vector of features)

	freqA	freqC	freqG	freqT	freqAA
1	0.26406926	0.24675325	0.2597403	0.22943723	0.100000000
2	0.26666667	0.24166667	0.2416667	0.25000000	0.109243697
3	0.29303107	0.20990764	0.2183039	0.27875735	0.088235294
4	0.25000000	0.21296296	0.2314815	0.30555556	0.074766355
5	0.38596491	0.28070175	0.1228070	0.21052632	0.142857143
6	0.23809524	0.33333333	0.1619048	0.26666667	0.028846154
7	0.26666667	0.14074074	0.2296296	0.36296296	0.074626866

Total 104 features

- frequencies of mono-, di- and trinucleotides (84 features)
- nucleotide correlation factors [3] (6 features)
- length and log(length)
- category-position-frequency (CPF) [4] (12 features)

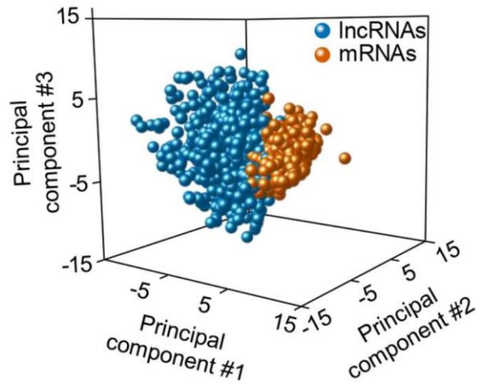


Fig. Features after vectorization separate lncRNAs and mRNAs in space of principle components. For each class of RNA molecules, only 1000 ORF candodates were randomly selected for compact representation.

Classification

Classification of the sequences was done by a classical **random forest** from *randomForest* R-package with 500 trees. The accuracy on the train set reached 98.3%. On the test dataset, the approach allowed identifying the true ORFs with an accuracy of 94.9%. In fact, 91.9% of ORFs that were identified in Ensembl human mRNA molecules demonstrated probability 0.9 or higher to be coding. Next, we estimated significance of the features by the Gini index. We identified some CPFs (HSS, HMK, HKK, see [4]) and the length as the most important features for determination of ORFs.

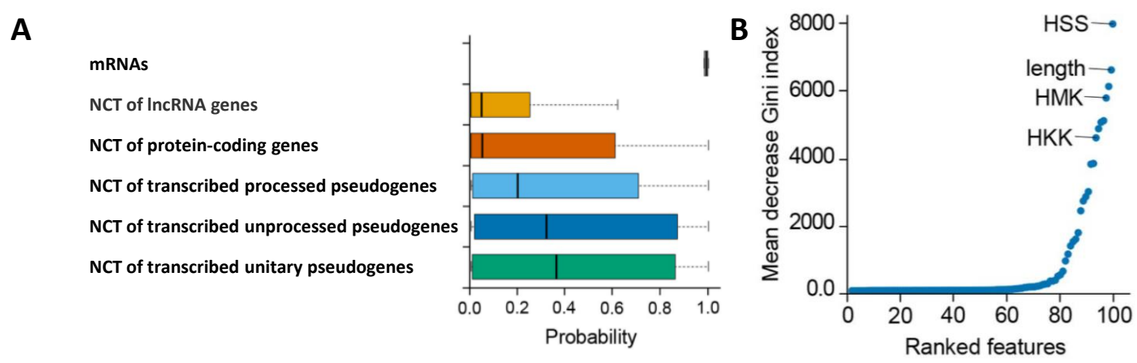


Fig. (A) Boxplot demonstrating the distribution of probability values for ORF candidates that were identified in human mRNAs and non-coding transcripts (NCT) in test dataset. (B) The importance of sequence features in determining the class of ORF candidates. The HSS, HMK and HKK features are the examples of CPF model parameters.

Annotation

Implemented annotation includes:

- length of 5'UTRs
- type of start codon
- start coordinate of open reading frames
- stop coordinate of open reading frames
- type of stop codon
- premature termination codon status of stop codon
- length of open reading frames
- length of 3'UTRs
- molecular weight of *in silico* predicted protein
- isoelectic point of a protein sequence
- potential protein interaction index

ORFhunteR Bioconductor package structure

Conclusions

ORFhunteR implemented the efficient computational approach for the identification of unknown ORFs in mRNA molecules based on vectorization of the sequence features of ORFs candidates and predicting the most evident by a random forest classifier. Accuracy of ORFs prediction 98.3% and 94.9% on the discovery and test data sets.

Developed approach has advantages over the competing modern strategies:

i) requires less computing resources and works much faster than neural nets;
ii) it is less prone to overfitting and uses a limited set of vectorized features (unlike the statistical approaches utilizing thousands of features);
iii) random forest classifier shows nice interpretability;
iv) parallel calculation of large data sets.

Availability. ORFhunteR is available at GitHub repository (https://github.com/rfctbiobsu/ORFhunteR), from Bioconductor and as a web application (http://orfhunter.bsu.by).

Funding

VVG, MMY, VVS and MKC were supported by the Ministry of Education of the Republic of Belarus, grant GPNI "Convergenciya–2020" N3.08.3 (registration number 20190531). PVN and MKC were supported by the Luxembourg National Research Fund (C17/BM/11664971/DEMICS).

References

1. Mardis, E.R. Nat Protoc 2017;12(2):213-218.
2. V.V. Grinev et al. bioRxiv 2021.02.05.429963.
3. Mao, R., et al. PLoS One 2014;9(8):e104049.
4. Bao, J., Yuan, R. and Bao, Z. BMC Bioinformatics 2014;15:321.

www.bsu.by
www.lih.lu